

- Hide menu
- Week 4: Appearance Matching
- Ungraded Plugin: 4.1
- Overview of Appearance Matching
- 4 min
- Ungraded Plugin: 4.2
- Shapes vs. Appearance
- 6 min
- Practice Assignment: 4.2
- Shapes vs. Appearance Self-Check Quiz
- Submitted
- Ungraded Plugin: 4.3
- Learning Appearance
- 6 min
- Practice Assignment: 4.3
- Learning Appearance Self-Check Quiz
- Submitted
- Ungraded Plugin: 4.4
- Principal Components Analysis
- 17 min
- Practice Assignment: 4.4
- Principal Components Analysis Self-Check Quiz
- Submitted
- Ungraded Plugin: 4.5
- Finding Principal Components
- 8 min
- Practice Assignment: 4.5
- Finding Principal Components Self-Check Quiz
- Submitted
- Ungraded Plugin: 4.6 PCA and SVD
- 7 min
- Practice Assignment: 4.6
- PCA and SVD Self-Check Quiz
- 3h
- Ungraded Plugin: 4.7
- Parameters Appearance Representation
- 12 min
- Practice Assignment: 4.7
- Parameters Appearance Representation Self-Check Quiz
- 3h
- Ungraded Plugin: 4.8
- Appearance Matching
- 14 min
- Practice Assignment: 4.8
- Appearance Matching Self-Check Quiz
- 3h
- Discussion Prompt: Week 4
- 4 Smartphones using 3D Sensors
- 10 min
- Week 4 Quiz

Your grade: 100%

Your latest: 100% • Your highest: 100% • Your lowest: 100% • Your average: 100%

Next item: →

4.5 Finding Principal Components Self-Check Quiz

1. For finding the Principal Component, the condition is defined as $\max(e^T R e) = 1$, such that $e^T e = 1$. 1 / 3 point

What does the condition $e^T e = 1$ ensure about e ?

- ☐ Orthogonal basis
- ☐ Radial basis
- ☒ Orthonormal basis
- ☐ None of the above

COUCH

Ready to review what you've learned before starting the assignment? I'm here to help.

- Help me practice
- Let's chat

Correct

Assignment details

The condition ensures that the eigenvector e_j is orthogonal to the eigenvector e_i for every $j \neq i$. It also ensures that each e_i is a unit vector, making it an orthonormal basis.

Aug 15, 2:36 AM IST

Unlimited

Try again

Your grade

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

100%



Like



Dislike



Report an issue